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Communication

Cycloadditions of 1-iminylphosphirane complexes with allenes

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ABSTRACT

A cascade carbonylative ring expansion and [2 + 2]/[4 + 2] cycloaddition of strained 1-iminylphosphirane complexes with aryl allenes were reported. The carbonylative ring expansion of 1-iminylphosphirane complexes provides an azaphosphacyclohexone complex intermediate with a C=P double bond. The following [2 + 2] or dearomatic [4 + 2] cycloaddition of this intermediate with allenes is modulated by the aryl substituents on the imino carbon. The regioselective [2 + 2] cycloaddition with 1,1-diarylallene provides an entry to bicyclo[4.2.0]octan-4-one skeletons featuring a four-membered phosphacyclobutane moiety. While dearomatic [4 + 2] cycloaddition was preferred with less aromatic naphthalene and yielded octahydrochrysene skeleton containing heteroatoms.

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Strained three-membered rings such as cyclopropanes, oxiranes and aziridines have been widely used as key building blocks for the synthesis of various high-value medium-sized rings or peri-fused ring systems [1]. However, the development of a similar chemistry with phosphiranes was somewhat hampered until now [2]. Based on our continuous interest on the phosphorus heterocycle chemistry [3], we demonstrated that 1-acylphosphirane tungsten complexes rearrange into 1,3-oxaphospholene complexes with a highly reactive P=C double bond, which is a phosphorus analogue of the Cloke-Wilson rearrangement (Scheme 1a) [4]. Recently, replacing the acyl group with an imino group led us to disclose a novel rearrangement of 1-iminylphosphirane complexes under a CO atmosphere. The successive cycloaddition with olefins yielded 7-aza-1-phosphanorbornane complexes (Scheme 1b) [5]. More interestingly, in the presence of $M(CO)_6$ (M = W, Mo), the reactions of 1-iminylphosphirane complexes and alkenes provide annulated heterodecalins with high molecular complexity and skeletal diversity (Scheme 1c). The proposed mechanism includes the carbonylative ring expansion and dearomatic Diels-Alder reaction [6].

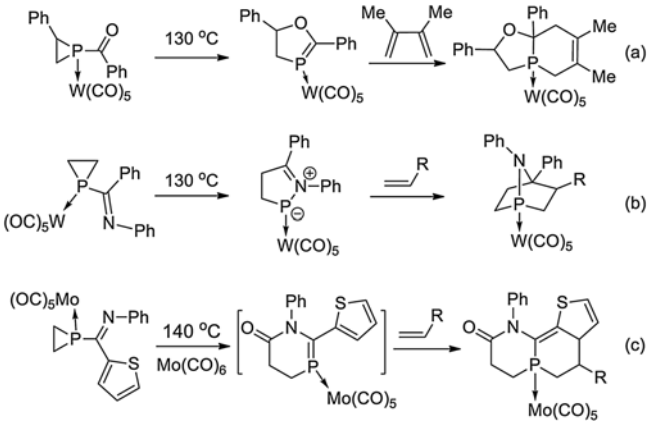
The carbonylative ring expansion of strained 1-iminylphosphirane complexes provides an azaphosphacyclohexone complex intermediate with a relatively stable C=P double bond. It is known that the highly reactive phosphaalkenes can give head-to-head [2 + 2] dimmers [7]. On the other hand, cumulated double bonds impart an extra reactivity to allene, making it a remarkably active component participating in a variety of cycloaddition reactions [8]. For example, the [2 + 2] cycloaddition reactions of allenes with imines afford azetidines, *via* a stepwise diradical mechanism rather than a concerted one-step mechanism [9]. This prompted us to investigate the reaction behavior of azaphosphacyclohexone complex with allenes.

We initiated our studies by investigating the reaction of 1-iminylphosphirane **1a** with 1,1-diphenylallene **2a** in the presence of $Mo(CO)_6$. As shown in Scheme 2, three kinds of products were isolated. The structure of **3aa** was unambiguously established by the X-ray crystallographic analysis (Fig. 1). The phosphorus atom attached to the central carbon atom of the allene. The strain within the 4-membered ring is obvious when looking at the long P1-C9 bond at 1.91 Å. The structure of **4aa** was established by comparing the NMR data of similar structure depicted in Scheme 1c, and was identified by a characteristic = CH_2 resonance in the ^{135}dept NMR spectrum at δ = +131.7 (J_{CP} = 21.2 Hz). Comparison of the NMR spectra of **4aa** and **5aa**, the disappearance of characteristic = CH_2 resonance and appearance of a new sp^3 CH_2 resonance at high-field indicated the formation of **5aa**. Similarly, replacing the thienyl group in imino moiety (**1a**) by 2-naphthyl (**1b**) provided corresponding product **3ba** along with **6ba**.

To understand the reaction mechanism, the reaction was performed without 1,1-diphenylallene **2a**. The ^{31}P NMR of crude reaction solution shows a new peak at +132 ppm which indicate the formation of intermediate **7** reference to the literature report [6]. Subsequent reaction of **7** with **2a** led to the formation of the expected **3ba** and **6ba**. Thus, a cascade mechanism for this reaction

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Scheme 1. Important ring expansions of phosphirane complexes.

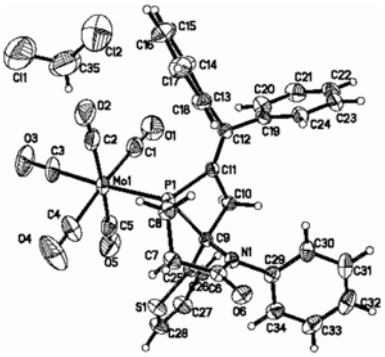
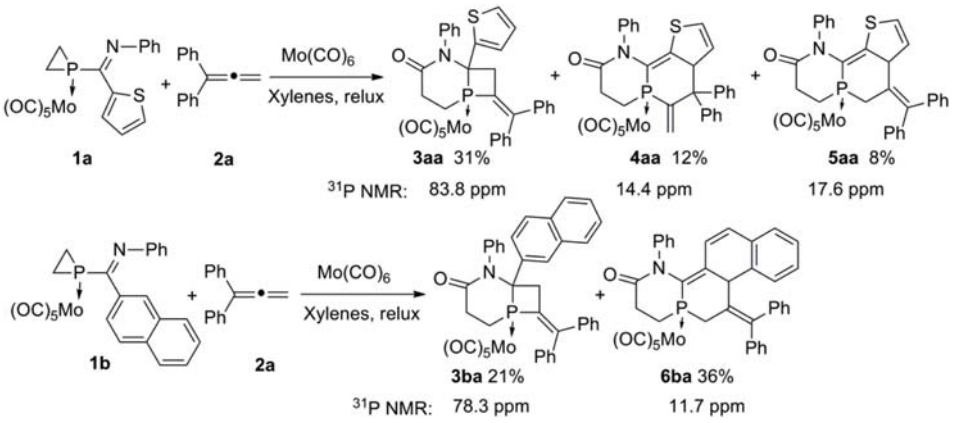


Fig. 1. Molecular structure of diphosphane **3aa**. Selected distances [Å] and angles [°]: P1-C8 1.833(4), P1-C9 1.909(3), P1-C11 1.822(2), C9-C10 1.564(4), C10-C11 1.511(5), C11-C12 1.330(5), P1-Mo1 2.4930(9). C8-P1-C9 103.55(16), C8-P1-C11 111.04(18), C9-P1-C11 76.51(15).

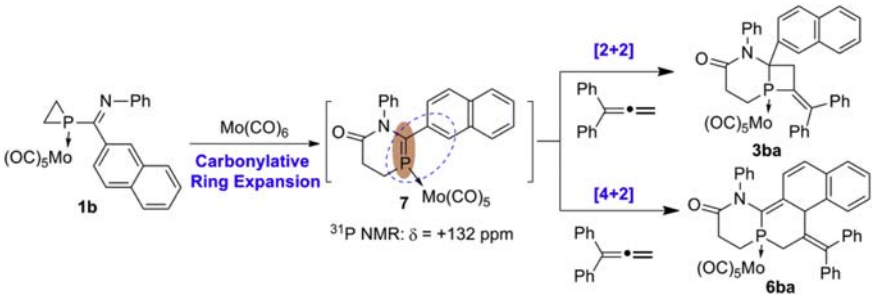
was proposed (Scheme 3). The carbonylative ring expansion of 1-iminylphosphirane complexes results in the formation of a reactive azaphosphacyclohexone derivative **7**. When **7** undergoes [2 + 2] cycloaddition with the terminal double bond of allene **2a**, the phosphacyclobutane fused **3ba** is produced. The good regio-selectivity of [2 + 2] cycloaddition to form **3ba** probably due to the steric hindrance of two phenyl groups inhibited the reactivity of the internal double bond of allene. While, the interesting dearomatic [4 + 2] cycloaddition [10] of **7** with **2a** gives peri-fused heterocycles **6ba**.

Replacing thiophene and naphthalene substituents with benzenoid resulted in the selective formation of the [2 + 2] cycloadducts **3** (Table 1). These results indicated that the selectivity between [2 + 2] and dearomatic [4 + 2] cycloaddition reactions might be tuned by the aryl substituent on imino carbon. The reaction with **1c** gave **3ca** as a sole product in moderate yield (Table 1, entry 1). The efficiency was found to be related to the electron richness of the aryl group on imino carbon, a general trend followed the order electron rich (**1d**, entry 2, 55%) > neutral (**1c**, entry 1, 48%) > electron deficient (**1e**, entry 3, 35%). Electron donating group increases the electron density of C=P double bond in the intermediates which promote the activities of the C=P double bond towards to allenes. Both electron-rich (Table 1, entry 4) and electron-deficient (Table 1, entry 5) aryl substituted allenes worked smoothly with **1c** and afford the corresponding [2 + 2] cycloadducts **3cb** and **3cc**. These [2 + 2] cycloadditions with 1, 1-diarylallenes proceeded exclusively at the less substituted double bond. When a less steric hindered 1-phenylallene **2d** was employed, both double bonds in allenes could participate in the [2 + 2] cycloaddition and yielded a pair of regio-isomers (Table 1, entries 6 and 7). All the new P—C bonds were selectively formed with the central carbon of allenes. Unfortunately, this reaction did not proceed with electron-poor allenoate (Table 1, entry 8).

The result with 1-phenylallene **2d** promoted us to revisit the reaction behaviors of thienyl and naphthyl substituted **1a** and **1b** with the less steric hindered allenes. To our delight, the decreased steric hindrance of allene improved both the chemical- and regio-selectivities. The dearomatic [4 + 2] cycloadditions, which involving the less aromatic thienyl and naphthyl moieties, became more facile than the competing [2 + 2] cycloadditions (Scheme 4). Moreover, the [4 + 2] reaction with the internal C=C double bond is preferred over the terminal ones of the allene. In all these cases, the products were purified as a mixture of [4 + 2] cycloadducts and the ratios of two adducts were determined by ^{31}P NMR. The structure of major product **6bd** was fully confirmed by single-crystal X-ray crystallography analysis (Fig. 2). The X-ray structure of **6bd** confirmed the correction of structure **4aa**. It is worth noting that the regioselectivity decreased with an electron-rich aryl-substituted allene **2g** (3:1), which indicated the regioselectivity was ascribed to the electronic and steric effects

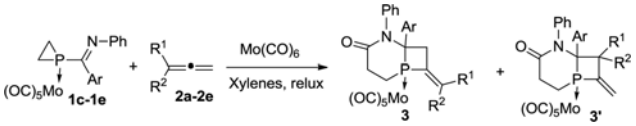


Scheme 2. Reactions of 1-iminylphosphirane complex **1a** and **1b** with 1,1-diphenylallene **2a**.



Scheme 3. Plausible reaction pathway.

Table 1
Reaction of phosphirane complexes **1c**-**1e** with allenes **2a**-**2e**.



Entry	Phosphirane		Allene			Product	Yiled (%)[a]
	1	Ar	**2**	R^1	R^2		
1	**1c**	Ph	**2a**	Ph	Ph	**3ca**	48
2	**1d**	*p*-MeOPh	**2a**	Ph	Ph	**3da**	55
3	**1e**	*p*-CF_3Ph	**2a**	Ph	Ph	**3ea**	35
4	**1c**	Ph	**2b**	*p*-MeOPh	*p*-MeOPh	**3cb**	61
5	**1c**	Ph	**2c**	*p*-FPh	*p*-FPh	**3cc**	53
6	**1c**	Ph	**2d**	Ph	H	**3cd+3cd'**	43[b]
7	**1d**	*p*-MeOPh	**2d**	Ph	H	**3dd+3dd'**	61[b]
8	**1c**	Ph	**2e**	CO_2Et	H		0

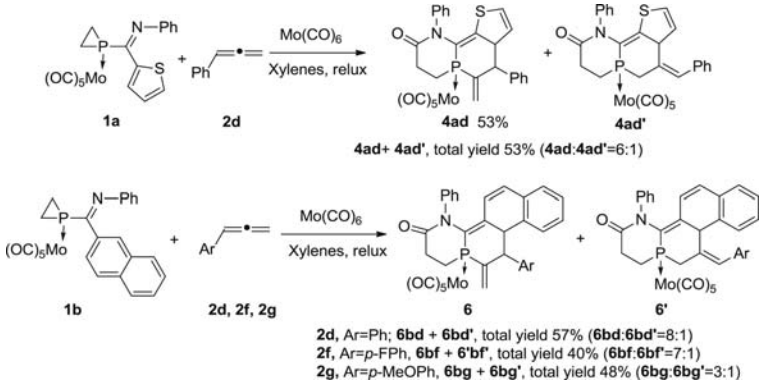
[a] Yields of the isolated product.
[b] A mixture of **3** and **3'** in a ratio of 1:1, the ratios were determined by ^{31}P NMR spectra.

of allene, but the steric hindrance has a more pronounced impact. It should be note that the reaction with naphthalene substituted **1b** provides an access to the phospha-hydrochrysene skeleton [11].

In summary, we describe a cascade carbonylative ring expansion and [2 + 2]/[4 + 2] cycloaddition of 1-iminylphosphirane complexes with aryl allenes. Particularly, [2 + 2]/[4 + 2] cycloaddition in the second step could be modulated by the aryl substituents on the imino carbon of phosphirane complexes. High aromatic benzenoid improved the [2 + 2] cycloaddition selectivity and providing a practical entry to a variety of bicyclo [4.2.0]octan-4-one skeletons featuring phosphacyclobut-ane rings. While dearomatic [4 + 2] cycloaddition was preferred with naphthalene and yielded octahydrochrysene skeleton containing heteroatoms.

Declaration of competing interest

The authors declare that there are no conflicts of interest.



Scheme 4. Reactions of 1-iminylphosphirane complex **1a** and **1b** with 1-arylallene **2**.

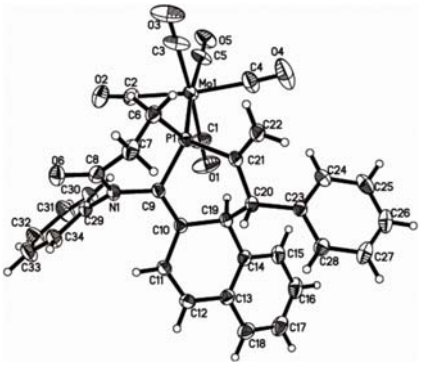


Fig. 2. Molecular structure of diphosphane **6bd**. Selected distances [Å] and angles [°]: P1-C6 1.839(4), P1-C9 1.820(3)(3), P1-C21 1.841(4), C21-C22 1.314(5), C8-O6 1.212(5), P1-Mo1 2.5104(12). C6-P1-C21 103.48(18), C6-P1-C9 98.04(19), C9-P1-C21 97.31(16).

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.cclet.2020.05.026.

References

[1] (a) T.F. Schneider, J. Kaschel, D.B. Werz, Angew. Chem. Int. Ed. 53 (2014) 5504–5523;
(b) Y. Wang, Z.X. Yu, Acc. Chem. Res. 48 (2015) 2288–2296;
(c) C. Ebner, E.M. Carreira, Chem. Rev. 117 (2017) 11651–11679;
(d) G. Fumagalli, S. Stanton, J.F. Bower, Chem. Rev. 117 (2017) 9404–9432.
[2] (a) F. Mathey, Chem. Rev. 90 (1990) 997–1025;
(b) R.E. Bulo, H. Jansen, A.W. Ehlers, et al., Angew. Chem. Int. Ed. 43 (2004) 714–717.
[3] (a) J. Wong, Y.X. Li, Y.W. Hao, et al., Angew. Chem. Int. Ed. 54 (2015) 12891–12893;
(b) Y.W. Hao, C.L. Zhang, Y.B. Mei, et al., Dalton Trans. 45 (2016) 8284–8290;
(c) Z.S. Hu, Z.Y. Li, K. Zhao, et al., Org. Lett. 17 (2015) 3518–3520;
(d) H.Y. Huang, G.Y. Tao, Z.B. Wei, et al., Organometallics 37 (2018) 464–468;
(e) H.Y. Huang, H.Y. Luo, G.Y. Tao, et al., Org. Lett. 20 (2018) 1027–1030;
(f) M. Wang, Y. Xu, D.H. Wei, et al., Chem. Eur. J. 25 (2019) 15036–15039;
(g) Y.Y. Song, L.L. Wang, Z. Duan, F. Mathey, Chin. Chem. Lett. 31 (2020) 329–332.
[4] R.Q. Tian, C.L. Zhang, Y. Xu, et al., Chem. Eur. J. 23 (2017) 13006–13009.
[5] Y. Xu, M. Wang, D.H. Wei, et al., Dalton Trans. 48 (2019) 5523–5526.
[6] Y. Xu, C.Y. Jiao, J. Li, et al., Org. Lett. 21 (2019) 9512–9515.
[7] S. Wang, K. Samedov, S.C. Serin, et al., Eur. J. Inorg. Chem. 26 (2016) 4144–4151.
[8] (a) N. Krause, A.S.K. Hashmi, Modern Allene Chemistry, Vols. 1-2, Wiley-VCH Verlag GmbH & Co., Weinheim, 2004;
(b) M.R. Siebert, J.M. Osbourn, K.M. Brummond, et al., J. Am. Chem. Soc. 132 (2010) 11952–11966;
(c) T. Iwata, F. Inagaki, C. Mukai, Angew. Chem. Int. Ed. 52 (2013) 11138–11142;
(d) F. Lopez, J.L. Mascarenas, Chem. Soc. Rev. 43 (2014) 2904–2915;
(e) W. Ding, N. Yoshikai, Angew. Chem. Int. Ed. 58 (2019) 2500–2504;
(f) X.L. Han, P.P. Lin, Q.J. Li, Chin. Chem. Lett. 30 (2019) 1495–1502.
[9] E. Schaumann, H. Mrotzek, Tetrahedron 35 (1979) 1965–1968.
[10] (a) A.R. Pape, K.P. Kaliappan, E.P. Kündig, Chem. Rev. 100 (2000) 2917–2940;
(b) S.H. Meiere, J.M. Keane, T.B. Gunnoe, et al., J. Am. Chem. Soc. 125 (2003) 2024–2025;
(c) B.K. Liebov, W.D. Harman, Chem. Rev. 117 (2017) 13721–13755.
[11] X.Q. Zhang, T. Guzi, L. Pettus, A.G. Schultz, Tetrahedron Lett. 43 (2002) 7605–7608 and references cited therein.